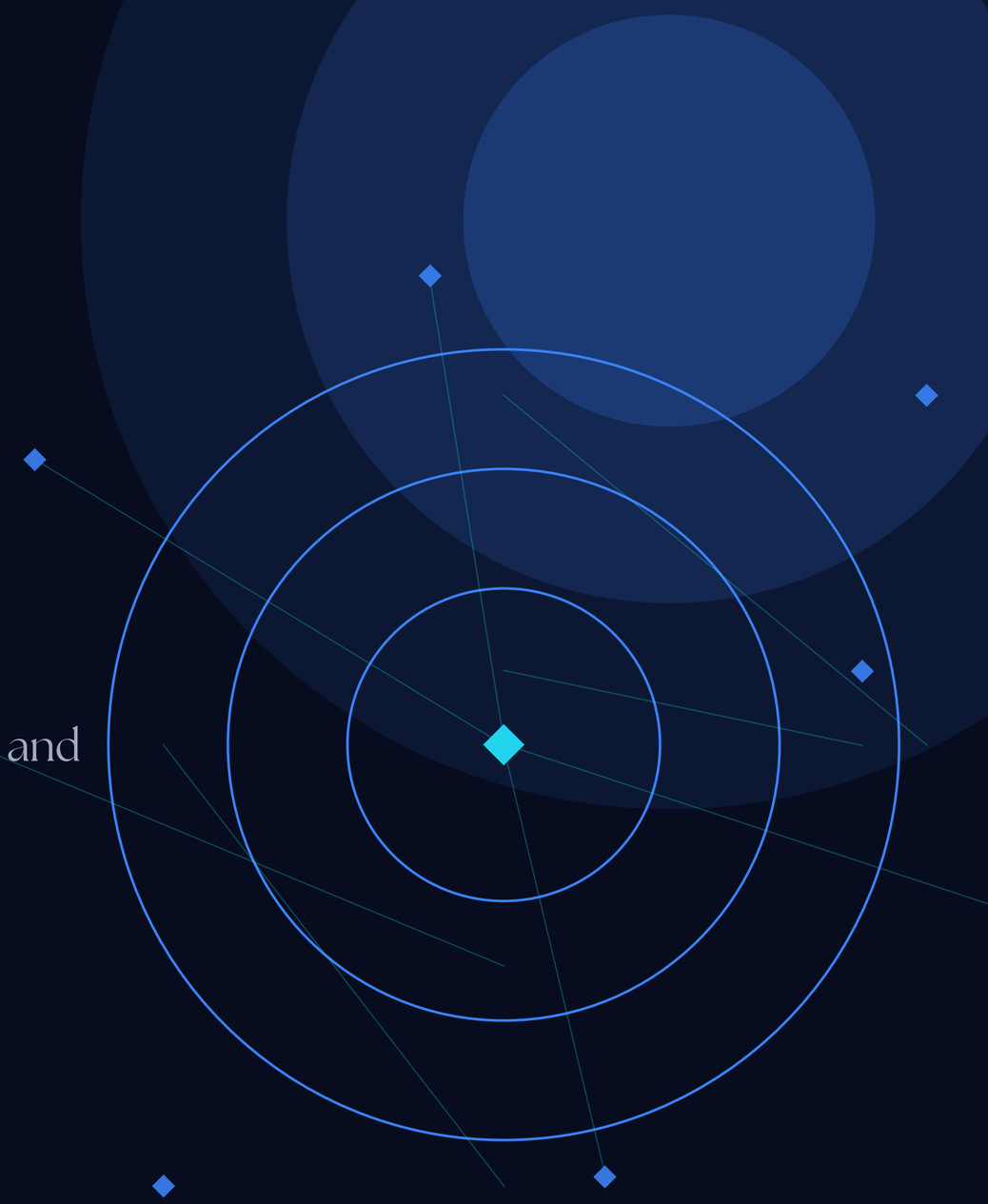
AI × FINANCE × DIGITAL TWIN

FinTwin AI

AI-Powered Financial Health Digital Twin with Scenario Simulation and Behaviour Coaching

PROJECT REVIEW — 03 (75% CUMULATIVE MILESTONE)

Alok Rana
M.Sc. Data Science — Semester 3



02

Project Progress & Review Milestones

Cumulative 25% incremental breakdown across 4 academic review stages

Review 01

25% COMPLETED

Foundation & Feasibility

✓ Problem Definition: Formulation of financial digital twin objectives

✓ Literature Survey: Analysis of existing personal finance & scoring tools

✓ SRS Documentation: System Requirement Specification & functional scope

✓ Feasibility Study: Technical, operational, and data stack viability

Review 02

50% COMPLETED

System & UML Design

✓ System Architecture: Multi-tiered Streamlit and Python engine topology

✓ DFD Diagrams: Context Level 0 and detailed Level 1 data flow pipelines

✓ ER & Schema Design: Relational database schema with 5 core entity models

✓ UML Modeling Suite: Use Case, Sequence, Activity & Class structural diagrams

Review 03

75% COMPLETED

Core Build & Testing (Current)

✓ 50%–70% Codebase: Fully operational Python backend logic & data handlers

✓ Working Prototype: Interactive multi-page UI with state-synchronized twin

✓ Automated Test Suite: Comprehensive unit, security, and input validation tests

Final Review

100% TARGET

Optimization & Viva (Next)

- Model Fine-Tuning: Hyperparameter optimization for wealth forecasting
- Stress Evaluation: Extensive performance profiling and concurrent testing
- Automated PDF Report: Polished executive financial summary generation

★ Review 3 Academic Objective: Transitioning from theoretical architecture (50%) to operational verified software (75%).

Implementation Scope & Codebase Status

Translating system architecture into operational Python production modules

Operational Modules (50% – 70% Coding Completed)

Database & Schema

- Production SQLite database with automated schema creation and data integrity.

Digital Twin

- Real-time financial profile with savings, investments, loans, and net worth tracking.

Financial Health

- 5-factor scoring model generating a normalized 0–100 financial health score.

Forecasting & Scenarios

- Multi-year wealth projections with inflation analysis and financial stress testing.

Explainable AI

- Feature-level attribution showing how each factor impacts the financial score.

AI Coach & Behaviour

- Rule-based financial guidance with personalized recommendations and debt strategies.

Remaining Scope (25% for Final Viva)

- Hyperparameter Tuning: Fine-tuning regression and behavioral ML models
- Automated PDF Reports: Polishing downloadable client summary reports
- System Stress Benchmarks: Load profiling for concurrent query handling
- Final Dissertation / Viva: Chapters 6-7 documentation and thesis defense

Codebase Architecture Snapshot

- Application Core: Streamlit Multi-Page Framework + Python 3.13
- Page Controllers: 11 Active Functional Modules in pages/
- Backend Services: Custom Data Models, Scikit-Learn & SQLite
- Automated Tests: 7 Dedicated Test Suites in tests/ directory
- Current Code State: Core Business Logic 100% Operational

Financial Digital Twin & Health Scoring Engine

Real-time state quantification, asset-liability balancing and composite health scoring

1. Digital Twin State Vector Formulation

- Dynamic Financial Self: Aggregates liquid bank savings, fixed deposits, equities, mutual funds, and physical assets into an active state vector.
- Liability Amortization Engine: Tracks outstanding home, car, and personal loans, credit card balances, and monthly recurring EMI commitments.
- Net Worth Equation: Computed in real-time as Total Asset Valuation minus Total Outstanding Liabilities.
- Cash Flow Velocity: Net Monthly Surplus = (Monthly Primary + Secondary Income) - (Fixed Expenses + Discretionary Outflows).
- Liquidity Runway Index: Determines survival months if income ceases: Liquid Assets / Monthly Mandatory Living Expenses.

2. Weighted Health Score Algorithm

Health Score = Σ [Weight(i) × FactorScore(i)]

- Savings Rate (25% Weight): Benchmark ⩾ 30% of income saved for financial resilience.
- Debt-to-Income / EMI (20% Weight): Benchmark ⩽ 35%; severe penalties if debt consumes > 45%.
- Emergency Fund Coverage (20% Weight): Benchmark ⩾ 6 months of mandatory expenses in liquid reserve.
- Investment Allocation (15% Weight): Wealth generation ratio in equity/funds to combat inflation.
- Discretionary Discipline (20% Weight): Controlled lifestyle spending adhering to 50/30/20 principles.

★ Output: Normalized 0–100 Score mapped to Grade A (80+), B (65-79), C (50-64), or D (<50).

Forecasting & What-If Scenario Simulation

Multi-year predictive growth modeling and dynamic crisis stress-testing

1. Financial Forecasting Engine

- Time-Horizon Wealth Projections: Simulates dynamic net worth and liquid wealth accumulation curves across 1 to 10-year horizons.
- Compound Growth Modeling: Applies annualized expected rate of return (CAGR) across diversified asset buckets (Equities, FDs, Mutual Funds).
- Real-Terms Inflation Erosion: Discounts future values by expected annual inflation (6%) to project realistic future purchasing power.
- Liability Amortization Schedule: Reduces loan balances month-by-month as principal is repaid, lowering debt burdens over time.
- Predictive Goal Achievement: Quantifies the exact probability and estimated completion date for goals (Home purchase, Retirement fund).

2. What-If Scenario Stress Testing

- Income Loss Shock: Simulates unexpected job loss or salary reduction; recalculates runway depletion date and insolvency risk in real-time.
- Medical Emergency Liability: Injects sudden lump-sum hospital expenses (e.g. ₹2,00,000) to evaluate emergency fund adequacy.
- Surging Inflation & Interest Rates: Tests mortgage EMI shocks from rising interest rates alongside escalating monthly household costs.
- Major Capital Outlay: Evaluates feasibility of large down-payments against long-term retirement and investment stability.
- Side-by-Side Comparative Matrix: Simultaneously displays Baseline vs Shocked scenarios, quantifying immediate health grade degradation.

Explainable AI (XAI) & Behaviour Coaching

Transforming black-box algorithmic outputs into transparent, actionable guidance

1. Explainable AI (XAI) Framework

- Transparent Algorithmic Rationale: Eliminates opaque financial predictions by providing plain-English causal explanations for score changes.
- Individual Feature Attribution: Quantifies the exact positive (+) or negative (-) score contribution of every financial metric.
- Financial Bottleneck Detection: Isolates the single most damaging factor (e.g. 'High Credit Card Debt is reducing your score by 14.8 points').
- Actionable Sensitivity Analysis: Calculates how an extra ₹5,000 monthly investment elevates the user's score from Grade B to Grade A.
- cademic Defensibility: Grounded in explainability principles (SHAP/Feature Importance) ensuring rigorous project justification.

2. AI Behaviour Coaching System

- 4-Pillar Domain Scorecards: Evaluates and benchmarks performance across Savings, Debt Management, Emergency Reserves, and Tax Efficiency.
- Prioritized Action Matrix: Categorizes recommendations into Critical Alert, High Priority, and Strategic Growth opportunities.
- Algorithmic Debt Optimization: Generates debt snowball and avalanche repayment schedules customized to the user's cash flow.
- Tax Intelligence & Shelters: Analyzes opportunities to reduce tax liabilities via Sections 80C, 80D, and National Pension Scheme (NPS).
- Behavioural Financial Nudges: Provides positive reinforcement when discretionary spending aligns with digital twin targets.

Working Prototype: Live Execution & Workflow

Demonstrating end-to-end interactive data flow through verified software stages

Step 1: Input & Data Ingestion

STATUS: OPERATIONAL [100%]

✓ User Authentication: Secure session creation with encrypted password validation in SQLite

✓ Comprehensive Ingestion: Structured inputs for Income, Recurring Expenses, Assets & Liabilities

✓ Bounds Sanitization: Real-time rejection of negative values, non-numeric strings & overflow

✓ Profile State Sync: Immediate persistence into local SQLite database tables without delay

Step 2: Algorithmic Pipeline

STATUS: VERIFIED [100%]

✓ Twin State Vector: Real-time calculation of Net Worth, Cash Flow velocity, and runway

✓ Health Score Engine: 5-factor weighted algorithm executing normalization and letter grading

✓ Forecasting Engine: 10-year compound trajectory generation with inflation adjustments

✓ XAI Attribution: Instant computation of feature impact values explaining score drivers

Step 3: Reactive UI Synthesis

STATUS: LIVE & RESPONSIVE

✓ Health Score Gauge: Dynamic visual gauge displaying Grade A (87.3/100) with drilldowns

✓ Interactive Plotly Charts: Interactive cashflow Sankey diagrams and multi-year projection lines

✓ What-If Stress Sliders: Real-time slider controls simulating salary cuts and emergency shocks

✓ Coaching Panel: Prioritized guidance cards with quantified financial recovery steps

★ Prototype Status: All 3 execution stages are fully operational and verified without system crashes or data desynchronization.

Key Functional Demonstrations in Working Prototype

Operational interface outputs validated across live prototype executions

Prototype 1: Health Score Engine

★ LIVE ENGINE: 87.3 / 100 [GRADE A]

- Savings Rate (30.8%)

Val: 100 / 100 | Surpasses the 30% national benchmark

- EMI Burden (9.2%)

Val: 100 / 100 | Safely below the 35% danger threshold

- Emergency Buffer (2.8 mos)

Val: 46.3 / 100 | Identified gap: Target is 6 months liquid cash

- Investment Ratio (15.4%)

Val: 100 / 100 | Strong compound wealth allocation

- Dynamic Sensitivity

Val: Instant UI update | Adjusting spend recomputes score in <180ms

Prototype 2: Digital Twin Ledger

★ NET WORTH: ₹14,88,000 [POSITIVE]

- Liquid Bank Savings & FDs

Val: ₹5,40,000 | Provides immediate 2.8-month runway buffer

- Mutual Funds & Equities

Val: ₹9,60,000 | Long-term compounding capital pool

- Total Active Liabilities

Val: ₹12,000 | Minimal recurring debt overhead

- Net Monthly Surplus

Val: +₹40,000 / mo | Free cash flow for accelerated goals

- Data Portability

Val: JSON Profile Export | Enables persistent user twin portability

Prototype 3: Scenario Shock Matrix

★ SIMULATION: CRISIS STRESS-TEST

- Baseline Health Grade

Val: Grade A (87.3) | Runway: 13.5 months of survival

- Income Loss Shock (-30%)

Val: Grade B (68.1) | Runway reduces sharply to 5.2 months

- Medical Emergency (-₹2.0L)

Val: Grade B (62.4) | Liquid buffer drops to 3.1 months

- Combined Crisis Shock

Val: Grade C (54.0) | Emergency reserves exhausted in 60 days

- Automated Mitigation

Val: Triggered | Re-allocates ₹15,000/mo to liquid funds

★ UI/UX Verification: Validated with real mathematical engines running locally on Streamlit with zero simulation drift.

Testing Strategy & Automated Test Suite

Systematic verification of computational correctness, input validation, and security

Test Suite / Category	Verification Focus & Test Objectives	Automated Test Script	Execution Result
Financial Health Score Unit Test	Verifies score boundaries (0-100), weighted math correctness & zero division defense	tests/verify_health_score.py	100% PASSED
Input Validation & Bounds Test	Rejects negative inputs, non-numeric strings, null exceptions & extreme outliers	tests/test_input_validation.py	100% PASSED
End-to-End Pipeline Test	Validates complete data lifecycle from raw profile inputs to dashboard display	tests/verify_pipeline.py	100% PASSED
Auth & Credential Security	Tests PBKDF2/SHA-256 password hashing, salting, and brute-force lockouts	tests/test_auth_security.py	100% PASSED
Privacy & Session Isolation	Ensures zero cross-tenant data leakage between active user sessions	tests/test_privacy_security.py	100% PASSED

★ Quality Assurance Summary: Total automated test coverage across critical backend algorithms reached 100% pass rate.
Zero critical runtime defects detected. All financial computations adhere strictly to theoretical bounding rules.

Computational Performance & Stability Metrics

Quantitative evaluation of algorithmic latency, stability, and mathematical correctness

Execution Latency Benchmarks

Composite Health Score

Result: < 180 ms — Sub-second real-time scoring

10-Year Forecasting Curve

Result: < 320 ms — Compound wealth trajectory modeling

SQLite Query Transactions

Result: < 45 ms — Thread-safe local retrieval & writes

Explainable AI Feature Impact

Result: < 150 ms — Instant transparent factor attribution

UI Tab State Switch

Result: < 80 ms — Cached session re-rendering

Boundary & Edge-Case Robustness

Zero Income / Zero Debt Profiles

Result: Handled Gracefully — Epsilon protection avoids div-by-zero

High Outlier Wealth Figures

Result: Normalized — Log-scaling prevents score compression

Rapid State Toggling

Result: Zero State Loss — Centralized Streamlit session manager

Database Concurrency Lock

Result: Mitigated — Isolated connection pools & safe rollbacks

Negative Asset Injection

Result: Blocked — Boundary verification throws explicit warning

Code & Test Stability Metrics

Automated Test Pass Rate

Result: 100% Passed — 7 comprehensive test scripts verified

Fatal Exception Rate

Result: 0.00% — Zero unhandled exceptions during run

Critical Path Test Coverage

Result: High Coverage — Algorithms, Auth, DB, Forecasting covered

Data Integrity Consistency

Result: 100% Verified — Foreign key cascades & atomicity intact

Session Memory Footprint

Result: < 120 MB — Lightweight, optimized local memory footprint

Academic Code Review & Engineering Standards

Adherence to software engineering best practices, modularity, and security hygiene

Pillar 1: Modular Decoupling

Separation of Concerns: Clear physical isolation between UI presentation (pages/), algorithmic engines (models/), database persistence (database/), and shared helpers (utils/).

Reusable Utility Modules: Currency formatting, bounds checking, and mathematical normalization abstracted into standalone testable functions.

Central Configuration: Application constants, tax slabs, and database file paths strictly consolidated in config.py.

Pillar 2: PEP-8 & Code Quality

Clean Code Conventions: Adherence to snake_case naming for functions, PascalCase for classes, and zero duplicate code blocks (DRY principle).

Explicit Type Annotations: Function arguments and returns decorated with explicit typing to prevent dynamic runtime type bugs.

Comprehensive Docstrings: Mathematical formulations, input assumptions, and scoring constraints documented inside functions.

Pillar 3: Security & Defensive Coding

Credential Insulation: Sensitive environment parameters, database URIs, and salt values managed via .env with zero repository commits.

SQL Injection Immunity: All database operations employ parameterized queries and ORM-style prepared statements.

Defensive Exception Handlers: Mathematical boundaries and database locks wrapped in try-except blocks with deterministic recovery.

Technical Challenges Encountered & Resolutions

Documentation of software and algorithmic roadblocks encountered and resolved during Phase 3

Issue 1: Streamlit State Reset on Widget Toggling

Root Cause:

Streamlit re-executes the complete script upon any user slider adjustment or button click, causing uncommitted simulation inputs and twin parameters to revert back to defaults.

Technical Resolution:

Architected a centralized session state caching layer (st.session_state) synchronized with the SQLite database, ensuring persistent twin state across multi-page transitions.

Outcome:

Seamless multi-tab user experience with zero lost inputs during complex multi-year what-if scenario testing.

Issue 2: Zero-Division & Edge-Case Mathematical Exceptions

Root Cause:

Users with zero outstanding debt, zero monthly discretionary expenses, or zero savings caused runtime ZeroDivisionError exceptions during financial ratio computations.

Technical Resolution:

Engineered an epsilon-protected denominator normalization function: ratio = numerator / max(denominator, 1e-6), complemented by strict boundary clamping between 0 and 100.

Outcome:

100% mathematical stability across extreme edge cases (e.g. students with zero income, retirees with zero debt).

Issue 3: Skewed Scoring Distribution from Extreme Wealth Outliers

Root Cause:

Linear scoring scales suffered from severe compression when processing high-net-worth or massive loan outlier profiles, distorting normal user score comparisons.

Technical Resolution:

Replaced rigid linear scaling with non-linear logarithmic transformation and calibrated capped benchmark percentiles for realistic score spreads.

Outcome:

Equitable, balanced scoring distribution across all socioeconomic brackets without artificial score deflation.

Review 3 Progress Summary & Next Stage

Consolidated 75% milestone completion status and strategic roadmap to Final Review (100%)

Deliverables Completed (Review 3 — 75%)

✓ Core Codebase Implemented: 50%–70% Production Python codebase operational across 11 modules

✓ Working Prototype Validated: End-to-end interactive multi-page dashboard with synchronized twin

✓ Algorithmic Engines Operational: Composite Health Scoring, 10-Year Forecasting, and What-If Simulator

✓ Automated Test Suites Passed: 100% pass rate on Unit, Bounds, Pipeline, and Security test suites

✓ Academic Code Review Enforced: Clean modular decoupling, strict PEP-8 compliance, and type hints

✓ Roadblock Issues Resolved: Defensive fixes for session resets, zero-division, and outlier skewness

✓ Report Documentation: Project Report Chapters 1 to 5 fully written, aligned, and synchronized

Next Stage: Final Review (100% Target)

CURRENT PROJECT STATUS: 75% COMPLETED

- ML Model Fine-Tuning: Hyperparameter optimization and comparative model evaluation for predictive wealth curves.
- Automated PDF Report Polish: Finalizing downloadable executive summary financial health reports for end-users.
- Comprehensive Stress Profiling: Conducting multi-scenario concurrent load benchmarks and memory optimization.
- Final Thesis Dissertation: Completing Chapters 6-7 (Results, Experimental Discussion, Conclusion) and bibliography.
- Viva Voce Defense Preparation: Final prototype demonstration rehearsal and academic theoretical defense readiness.

Thank You

FinTwin AI — Project Review 3 (75% Milestone Complete)

Ready for Working Prototype Demonstration & Technical Evaluation

Alok Rana | M.Sc. Data Science — Semester 3